

FLOW AUTOMATE

Complete Platform Design & Implementation Guide

Node Library · Shape Designs · Listener Configurations
Loop Node · Wait Node · MCP Tool Connector
Field Types · Validations · Error Messages
Scheduler · Webhook · SOP Flows · JSON Data Contracts

Version 4.0 · For UI/UX Designers & Developers · May 2025

Section 1: Complete Node Shape Library

All 14 node shapes used on the Flow Automate canvas. Each shape has a distinct visual, colour, and port contract. Shapes marked NEW were added based on requirements for iteration, synchronisation, and time control.

1.1 Port & Connector Legend



1.2 Complete Shape & Colour Reference

Node	Shape · Colour	Ports	Rationale
Start	Filled circle · Green	0 in · 1 out	Universal play/go signal. One per flow.
End	Filled circle · Red	1 in · 0 out	Universal stop signal. Multiple allowed.
Human task	Stadium/pill · Amber	1 in · 1–3 out	Pauses flow for human input or approval.
Process / flow call	Rect + double side borders · Purple	1 in · 1 out + error	Invokes a saved flow as a sub-process.
Agent block	Dashed rounded rect · Purple	1 in · 1 out	AI agent — autonomous, non-deterministic.
Conditional	Diamond · Blue	1 in · True + False out	IF decision. Supports expression or AI prompt.
Router	Parallelogram · Teal	1 in · N named out	Routes to N branches. First-match or fan-out.
Connector / API call	Rect + thick left bar · Teal	1 in · Success + Error out	Outbound call — HTTP, pre-config, or MCP tool.
Default inputs	Rect + thick top bar · Gray	0–1 in · 1 out	Injects config, secrets, env vars into context.
Listener	Hexagon + antenna dot · Coral	0 in · 1 out	Always-on event source. Spawns flow per event.

Notification	Rect + bell icon · Amber	1 in · 1 out	Outbound alert — email, Slack, Teams, SMS.
Scheduler trigger	Dashed circle · Gray	0 in · 1 out	Per-flow cron/interval trigger (not shared).
Webhook trigger	Pentagon/flag · Purple	0 in · 1 out	Per-flow HTTP endpoint trigger (not shared).
Loop NEW	Rounded rect + loop icon · Pink	1 in · Body + Done out	Iteration — for-each, count, while, do-while.
Wait — Timer NEW	Horizontal hourglass · Orange	1 in · 1 out	Pauses flow for a duration or until a datetime.
Wait — Process Any NEW	Hourglass + N arrows · Orange	N in · 1 out	Fires when the FIRST of N branches arrives.
Wait — Process All NEW	Hourglass + N arrows (thick) · Orange	N in · 1 out	Fires only when ALL N branches have arrived.

Section 2: Node Definitions — Configuration, Validation & Errors

01 — START NODE



Start

Shape: Filled circle

Colour: Green

Ports: 0 in | 1 out

Description

Entry point of every flow. Exactly one Start node per flow. Fires on manual trigger, schedule, or external event. Passes an initial payload downstream.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Trigger type	enum	Required	Must be one of: manual schedule webhook event	trigger_type · enum
Label	string	Optional	Max 64 chars	name · string
SCHEDULE/WEBHOOK				
Cron expression	string	Conditional	Valid 5-field cron syntax. Regex: <code>^\(S+)\{4\}S+\$</code> Shown when trigger_type = schedule	cron_expression · string
Webhook secret	string(secret)	Conditional	Min 16 chars. Stored in vault. Shown when trigger_type = webhook	webhook_secret · vault_ref
Payload schema	json	Optional	Must be valid JSON Schema draft-07. Validated on save	initial_payload_schema · object
RELIABILITY				
Retry on fail	boolean	Optional	—	retry_on_fail · boolean · default: false
Timeout (s)	integer	Optional	Range: 1–86400. Error if exceeded	timeout_s · integer · default: 300

Error Messages

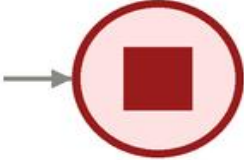
Error code	Trigger	UI message	Dev message	Recovery action
ERR_START_001	Duplicate Start node	Only one Start node is allowed per flow	flow_validation: multiple_start_nodes	Remove extra Start node
ERR_START_002	Invalid cron expression	Cron expression is not valid. Example: 0 9 * * 1-5	cron_parse_error: {expression}	Fix syntax using cron preview
ERR_START_003	HMAC verification failed	Incoming webhook could not be verified — check the secret	hmac_mismatch: expected {hash}	Verify caller HMAC signing

ERR_START_004	Payload schema mismatch	Incoming payload does not match the configured schema	schema_validation_error: {field} {reason}	Fix caller payload or relax schema
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Input / Output JSON

Input JSON <pre>{ "trigger": "webhook", "source": "siem_inbound" "timestamp": "2025-05-30T08:42:11Z" "metadata": { "hmac_verified": true } }</pre>	Output JSON <pre>{ "flow_id": "flow_siem_master_v2", "run_id": "run_7b4e91d" "triggered_by": "webhook", "started_at": "2025-05-30T08:42:11Z" }</pre>
Logic Applied Flow engine generates run_id (UUID v4) and stamps started_at. HMAC verified against vault secret. Failure exits immediately to failure End node.	

02 — END NODE



End

Shape: Filled circle

Colour: Red

Ports: 1 in | 0 out

Description

Terminal node. Multiple End nodes allowed for different exit paths. Sets run status and optionally emits a result webhook or triggers a downstream flow.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Label	string	Required	Max 64 chars. E.g. Success, Failure, Escalated	name · string
OUTPUT				
Result mapping	string	Optional	Must be valid JSONPath. Error if path not found at runtime	result_mapping · string · JSONPath
Emit result webhook	string(url)	Optional	Must be valid HTTPS URL	result_webhook_url · string
Trigger downstream flow	string(uuid)	Optional	Must be a valid published flow ID	downstream_flow_id · string

Notify on end	enum	Optional	Values: email slack none · default: none	notify_on_end · enum
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Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_END_001	No End node	Every flow must have at least one End node	flow_validation: missing_end_node	Add an End node
ERR_END_002	Unreachable End node	This End node cannot be reached from Start	flow_validation: unreachable_node:{id}	Connect to the flow graph
ERR_END_003	Result webhook failed	Could not POST result to the configured webhook URL	webhook_delivery_error: {url} status:{code}	Check URL availability

Input / Output JSON

Input JSON <pre>{ "ticket_id": "TK-2187", "result": { "sop_success": true } }</pre>	Output JSON <pre>{ "flow_id": "flow_siem_master_v2", "run_id": "run_7b4e91d" "status": "success", "duration_ms": 3331000 "audit_trail": { "me_audit_entry_id": "audit_4d8cla" } }</pre>
Logic Applied Assembles final run record. Stamps ended_at, computes duration_ms, writes to audit log.	

03 — HUMAN TASK NODE



Human task

Shape: Stadium / pill

Colour: Amber

Ports: 1 in | 1-3 out

Description

Pauses flow and assigns a task to a human operator. Three sub-types: User webform, Chat bot, Decision email/webpage. Flow resumes when the human responds.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string

SUB-TYPE				
Interaction mode	enum	Required	Values: webform chatbot decision_email	interaction_mode · enum
Assignee	string	Required	User ID, group name, or email. Max 256 chars	assignee · string
Task title	string	Required	Max 128 chars. Shown as subject / heading to human	task_title · string
Due in (minutes)	integer	Optional	Range: 1–43200 (30 days). Triggers escalation on expiry	due_in_minutes · integer · default: 60
SLA / ESCALATION				
Escalate to	string	Optional	User ID or group. Notified when SLA expires	escalate_to · string
Expiry action	enum	Optional	Values: escalate auto_approve auto_reject wait	on_expiry · enum · default: escalate
DECISION OPTIONS				
Decision options	string[]	Conditional	Min 1, max 8 options. Each max 64 chars. Required for decision_email mode	decision_options · string[]
One-click token expiry (h)	integer	Optional	Range: 1–168. Token becomes invalid after this time	token_expiry_h · integer · default: 24

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_HT_001	No assignee	Assignee is required. Enter a user ID, group name, or email	missing_required_field: assignee	Set assignee
ERR_HT_002	SLA exceeded — auto action	No response received within the SLA window. Auto-action taken: {action}	sla_expired: task_id:{id} action:{action}	Review escalation settings
ERR_HT_003	Invalid token	This approval link has expired or was already used	token_invalid_or_expired: {token_id}	Resend the task
ERR_HT_004	Assignee not found	The assignee '{value}' could not be resolved to a user or group	assignee_not_found: {assignee}	Check user directory

Input / Output JSON

Input JSON <pre>{ "assignee": "group:SOC_L1", "task_title": "Approve SOP for TK-2187", "interaction_mode": "decision_email", "due_in_minutes": 15, "decision_options": ["approve", "reject", "escalate_12"] }</pre>	Output JSON <pre>{ "response_type": "decision_email", "decision": "approve", "completed_by": "analyst_raj.kumar", "completed_at": "2025-05-30T08:49:03Z", "task_duration_s": 412 }</pre>
Logic Applied	

Tokenised one-click email sent to SOC_L1. SLA 15 min — auto-escalates on timeout. Analyst identity and timestamp merged into flow context for audit trail.

04 — PROCESS / FLOW CALL NODE



Description

Invokes another saved flow as a reusable sub-process synchronously or asynchronously. Secrets from the parent are never passed to the child.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
TARGET FLOW				
Target flow	string(uuid)	Required	Must be a valid published flow ID	sub_flow_id · string · uuid
Execution mode	enum	Required	Values: sync async · default: sync	execution_mode · enum
EXECUTION				
Input mapping	object	Optional	JSONPath source → child parameter name. Must be valid JSONPath expressions	input_mapping · object
Output mapping	object	Optional	JSONPath in child result → parent context key	output_mapping · object
Timeout (s)	integer	Optional	Range: 1–3600 · default: 300	timeout_s · integer
On child failure	enum	Optional	Values: fail_parent continue retry · default: fail_parent	on_child_failure · enum
Retry count	integer	Conditional	Range: 1–5. Required when on_child_failure = retry	retry_count · integer

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_PC_001	Target flow not found	The selected flow could not be found. It may have been deleted	sub_flow_not_found: {flow_id}	Select a valid published flow
ERR_PC_002	Child flow timeout	The sub-flow did not complete within the configured timeout	sub_flow_timeout: {flow_id} after:{timeout_s}s	Increase timeout or optimise child flow
ERR_PC_003	Child flow failed	The sub-flow ended with status: failed	sub_flow_failed: run_id:{run_id} reason:{reason}	Check child flow run log
ERR_PC_004	Input mapping error	JSONPath '{path}' did not resolve in the current context	input_mapping_error: path:{path}	Fix JSONPath or check upstream data

Input / Output JSON

Input JSON <pre>{ "sub_flow_id": "delta_inc_sop_flow", "wait_for_result": true "timeout_s": 300, "input_mapping": { "ticket_id": "TK-2187" "hostname": "WKST-042" } }</pre>	Output JSON <pre>{ "sub_run_id": "run_delta_c3f2a1", "status": "completed" "result": { "sop_success": true, "patch_kb": "KB5034441" } }</pre>
Logic Applied Parent calls child synchronously. Input mapping strips context to only required fields. Secrets never passed. sop_success: true returned with evidence URL.	

05 — AGENT BLOCK



Agent block
 Shape: Dashed rounded rect
 Dashed = non-deterministic — Purple
 Ports: 1 in | 1 out

Description

Runs an AI agent that autonomously reasons and executes multi-step tasks. Dashed border signals non-deterministic execution. Agent loop runs until goal met or step limit reached.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
MODEL & PROMPT				

Model	enum	Required	Values: gpt4o claude gemini custom	model · enum
System prompt	string	Required	Min 10 chars. Max 4000 chars. Supports {{variable}} injection	system_prompt · string
Goal	string	Required	Min 10 chars. Max 2000 chars. Supports {{variable}} injection	goal · string
TOOLS & LIMITS				
Tools enabled	string[]	Optional	Valid tool IDs from AutoX tool registry	tools_enabled · string[]
Max steps	integer	Required	Range: 1–50 · default: 10	max_steps · integer
Temperature	float	Optional	Range: 0.0–1.0 · default: 0.7	temperature · float
OUTPUT				
Output schema	json	Optional	Valid JSON Schema draft-07. Agent output validated and structured	output_schema · object
Confidence threshold	float	Optional	Range: 0.0–1.0. Below this triggers human escalation · default: 0.85	confidence_threshold · float
Memory	enum	Optional	Values: none session persistent · default: none	memory_mode · enum

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_AG_001	Max steps reached	The agent reached the maximum step limit without completing the goal	agent_max_steps: goal:{goal} steps:{max}	Increase max_steps or simplify goal
ERR_AG_002	Low confidence escalation	Agent confidence {value} is below threshold {threshold}. Escalating to human	agent_low_confidence: confidence:{val}	Review agent output or adjust threshold
ERR_AG_003	Model API error	The AI model returned an error: {reason}	model_api_error: model:{model} status:{code}	Retry or switch model
ERR_AG_004	Output schema mismatch	Agent output does not match the configured output schema	agent_output_schema_error: {field} {reason}	Adjust output schema or system prompt
ERR_AG_005	Tool execution failed	Tool '{tool_name}' returned an error during agent step {step}	agent_tool_error: tool:{name} step:{n}	Check tool availability

Input / Output JSON

Input JSON <pre>{ "goal": "Parse SIEM ticket, extract structured fields" "context": { "ticket_id": "TK-2187", "severity": "critical" }</pre>	Output JSON <pre>{ "client_name": "delta_inc", "severity_rank": 1 "incident": { "type": "malware", "subtype": "ransomware" }</pre>
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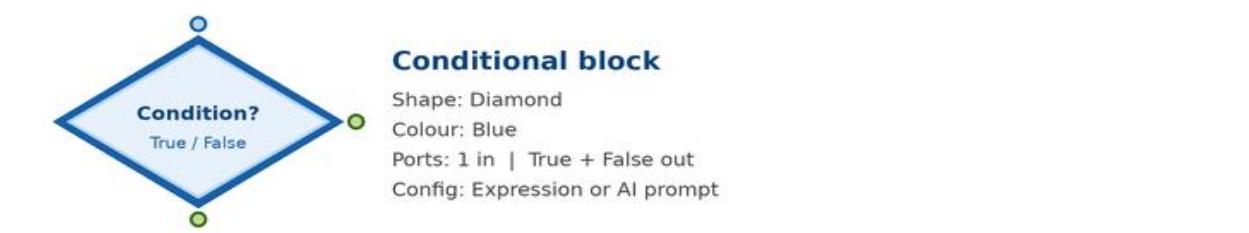
```
"max_steps": 1 }
```

```
"requires_human_approval": true,  
"confidence": 0.97 }
```

Logic Applied

Resolves CLIENT_B to delta_inc. Classifies ransomware. Sets requires_human_approval: true. Confidence 0.97 > 0.85 threshold — no escalation.

06 — CONDITIONAL BLOCK



Description

Evaluates a boolean expression against the JSON context and routes to True or False branch. Mode A: structured expression. Mode B: natural-language AI prompt. No data transformation — all fields pass through.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
CONDITION				
Condition type	enum	Required	Values: expression ai_prompt	condition_type · enum
Expression	string	Conditional	Valid JSONPath predicate. E.g. \$.severity_rank <= 1. Required when condition_type = expression	condition_expr · string
AI prompt	string	Conditional	Min 10 chars. Max 500 chars. Must be a yes/no question. Required when condition_type = ai_prompt	condition_prompt · string
BRANCHES				
True branch label	string	Optional	Max 64 chars · default: True	true_label · string
False branch label	string	Optional	Max 64 chars · default: False	false_label · string

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_CO_001	Invalid expression	The condition expression could not be parsed: {reason}	expression_parse_error: {expr}	Fix JSONPath syntax
ERR_CO_002	Expression field not found	Field '\$.{field}' does not exist in the current context	field_not_found: path:{path}	Check upstream node output
ERR_CO_003	AI prompt ambiguous	The AI prompt returned an uncertain result. Could not resolve to true or false	ai_condition_uncertain: prompt:{prompt}	Reword the prompt to be more specific
ERR_CO_004	No outgoing connection	The True or False branch has no connected node	flow_validation: dangling_branch:{branch}	Connect both branches

Input / Output JSON

Input JSON

```
{ "ticket_id": "TK-2187", "severity": "critical", "severity_rank": 1 }
```

Output JSON

```
{ "ticket_id": "TK-2187", "severity": "critical", "severity_rank": 1, "_branch": "true", "_condition": "severity_rank <= 1", "_evaluated_as": "1 <= 1 = true" }
```

Logic Applied

Condition evaluates 1 <= 1 = true. True branch fires Human task. All upstream fields pass through; _branch and _condition keys appended.

07 — ROUTER BLOCK



Router block

Shape: Parallelogram
Colour: Teal
Ports: 1 in | N named out

Description

Fans out to multiple named output paths based on a routing key. Unlike the binary Conditional, supports N branches. First-match mode or parallel fan-out.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
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IDENTITY				
Name	string	Required	Max 128 chars	name · string
ROUTING				
Routing key	string	Required	Must be a valid JSONPath. E.g. \$.client_id	routing_key · string · JSONPath
Routing mode	enum	Required	Values: first_match all_matching round_robin · default: first_match	routing_mode · enum
Routes	object[]	Required	Min 1 route. Each route must have a name (max 64 chars) and a condition expression	routes · array · min:1
Default route	string	Required	Must match one of the route names. Required as fallback	default_route · string
FAN-OUT LIMIT				
Max parallel branches	integer	Optional	Range: 1–50. Only applies when routing_mode = all_matching	max_concurrency · integer · default: unlimited

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_RO_001	No matching route	No route matched and no default route is configured	no_route_matched: key:{value}	Add a default route
ERR_RO_002	Routing key not found	The routing key '{field}' was not found in the context	routing_key_not_found: path:{path}	Fix JSONPath or check upstream data
ERR_RO_003	Duplicate route name	Two routes have the same name. Route names must be unique	duplicate_route_name: {name}	Rename one route
ERR_RO_004	Route condition error	Route '{name}' condition could not be evaluated: {reason}	route_condition_error: route:{name}	Fix route condition expression

Input / Output JSON

Input JSON <pre>{ "ticket_id": "TK-2187", "client_id": "CLIENT_B", "severity": "critical" }</pre>	Output JSON <pre>{ "ticket_id": "TK-2187", "client_id": "CLIENT_B", "_route": "CLIENT_B", "_route_label": "delta_inc_sop_flow", "_rule_index": 1, "_routing_mode": "first_match" }</pre>
Logic Applied Routing key: \$.client_id. Rule 1: CLIENT_B matches. First_match — stops evaluating. delta_inc_sop_flow used as child flow ID.	



Connector node

Shape: Rectangle + thick left bar
Colour: Teal / Green
Outbound API calls to external systems
Ports: 1 in | Success + Error out

Description

Makes an outbound call in one of three modes: HTTP/REST (raw), Pre-configured connector (named account), or MCP tool (AutoX MCP registry action). All three modes share the same shape — the mode selector at the top of the config panel drives all visible fields.

Mode A — HTTP / REST configuration

Field label	Field type	Required	Validation rule	Dev key · notes
MODE				
Connection mode	enum	Required	Values: http_rest connector mcp_tool	connection_mode · enum
HTTP / REST FIELDS				
Method	enum	Required	Values: GET POST PUT PATCH DELETE	method · enum
URL	string	Required	Must start with https://. Supports {{variable}}. Max 2048 chars	url · string · url
Headers	object	Optional	String→string map. Values support {{variable}}	headers · object
Body type	enum	Optional	Values: none json form_data raw	body_type · enum
Body content	string object	Conditional	Required when body_type != none. Supports {{variable}}	body · string or object
Auth type	enum	Optional	Values: none api_key bearer_token basic oauth2	auth_type · enum
API key value	string(secret)	Conditional	Min 8 chars. Stored in vault. Required when auth_type = api_key	api_key_value · vault_ref
Bearer token	string(secret)	Conditional	Min 8 chars. Required when auth_type = bearer_token	bearer_token · vault_ref
OAuth2 client ID	string	Conditional	Max 256 chars. Required when auth_type = oauth2	oauth2_client_id · string
OAuth2 client secret	string(secret)	Conditional	Min 8 chars. Required when auth_type = oauth2	oauth2_client_secret · vault_ref
RELIABILITY				
Timeout (ms)	integer	Optional	Range: 100–300000 · default: 5000	timeout_ms · integer
Retry count	integer	Optional	Range: 0–10 · default: 3	retry_count · integer

Success codes	<code>string</code>	Optional	Comma-separated ranges. E.g. 200-299,304 · default: 200-299	<code>success_codes</code> · <code>string</code>
Verify TLS	<code>boolean</code>	Optional	—	<code>verify_tls</code> · <code>boolean</code> · default: <code>true</code>
RESPONSE				
Response mapping	<code>object</code>	Optional	JSONPath → context key pairs. JSONPath must be valid	<code>response_mapping</code> · <code>object</code>
On error	<code>enum</code>	Optional	Values: <code>fail_flow continue retry</code> · default: <code>fail_flow</code>	<code>on_error</code> · <code>enum</code>

Mode C — MCP Tool configuration (NEW)



Connector — MCP tool mode

Shape: Rect + thick left bar (teal)
Purple badge: MCP tool mode
Select: server → tool → one action
Ports: 1 in · Success + Error out

Field label	Field type	Required	Validation rule	Dev key · notes
MCP SERVER				
MCP server	<code>string(uuid)</code>	Required	Must be a registered server in AutoX MCP registry with status: <code>online</code>	<code>mcp_server_id</code> · <code>string</code> · <code>uuid</code>
MCP TOOL				
Tool name	<code>string</code>	Required	Must be a tool published on the selected MCP server	<code>mcp_tool_name</code> · <code>string</code>
Action	<code>string</code>	Required	One action per node (by design). Must be valid action of selected tool	<code>mcp_action</code> · <code>string</code>
PARAMETERS				
Parameter builder	<code>object</code>	Required	Auto-generated from tool action schema. Values support <code>{{variable}}</code> . Required params must be filled	<code>mcp_params</code> · <code>object</code>
Context mapping	<code>object</code>	Optional	JSONPath → param name auto-suggestions. All mapped paths must be valid	<code>mcp_param_mapping</code> · <code>object</code>
AUTH				
Auth method	<code>enum</code>	Required	Values: <code>autox_managed api_key oauth2</code> · default: <code>autox_managed</code>	<code>mcp_auth_method</code> · <code>enum</code>
API key	<code>string(secret)</code>	Conditional	Min 8 chars. Required when <code>auth_method = api_key</code>	<code>mcp_api_key</code> · <code>vault_ref</code>
OAuth2 credential	<code>string(uuid)</code>	Conditional	Must reference a saved credential. Required when <code>auth_method = oauth2</code>	<code>mcp_oauth2_credential_id</code> · <code>uuid</code>

EXECUTION				
Timeout (ms)	integer	Optional	Range: 1000–300000 · default: 30000	timeout_ms · integer
Retry count	integer	Optional	Range: 0–5 · default: 2	retry_count · integer
RESPONSE				
Response mapping	object	Optional	JSONPath from tool response → flow context key	response_mapping · object
Full response key	string	Optional	Max 64 chars · default: mcp_result	response_key · string

Error Messages — all connector modes

Error code	Trigger	UI message	Dev message	Recovery action
ERR_CN_001	HTTP non-success status	The API call returned an unexpected status: {code}	http_error: url:{url} status:{code} body:{preview}	Check endpoint and auth
ERR_CN_002	Connection timeout	The request timed out after {timeout_ms}ms	connect_timeout: url:{url} after:{timeout_ms}ms	Increase timeout or check endpoint
ERR_CN_003	TLS verification failed	Could not verify the server certificate for {host}	tls_error: host:{host} reason:{reason}	Check cert or disable TLS verify
ERR_CN_004	MCP server offline	The MCP server '{name}' is currently offline	mcp_server_unavailable: server_id:{id}	Check server status in Integrations
ERR_CN_005	MCP tool not found	Tool '{tool}' was not found on server '{server}'	mcp_tool_not_found: tool:{t} server:{s}	Re-select tool from updated manifest
ERR_CN_006	MCP action failed	Action '{action}' returned an error: {reason}	mcp_action_error: tool:{t} action:{a} reason:{r}	Check params and server logs
ERR_CN_007	Response mapping error	JSONPath '{path}' did not resolve in the response	response_mapping_error: path:{path}	Fix JSONPath or check API response shape
ERR_CN_008	Retries exhausted	All {retry_count} retry attempts failed	max_retries_exceeded: url:{url} attempts:{n}	Review on_error policy or alert ops

Input JSON <pre>{ "method": "GET", "url": "https://me.internal/api/v3/tickets/new" "headers": { "X-API-Key": "....." } "query_params": { "status": "open", "limit": 1 } }</pre>	Output JSON <pre>{ "status_code": 200, "latency_ms": 187 "response": { "ticket_id": "TK-2187", "client_id": "CLIENT_B" "severity": "critical", "sop": { "sop_id": "SOP-EDR-001" } } "error": null, "mcp_meta": null }</pre>
Logic Applied HTTP/REST mode: GET to ME API. MCP mode: jsonrpc tool/call dispatched to AutoX MCP server. Response mapping extracts specific fields into flow context. mcp_meta block added for MCP calls with server_id, tool, action, duration_ms, retries.	

**Default inputs block**

Shape: Rectangle + thick top bar

Colour: Gray

Ports: 0-1 in | 1 out

Description

Injects static or environment-sourced values into the JSON context. Runtime values override defaults when the same key exists in both.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
VALUES				
Default values	object	Required	Must be valid JSON. Max 100 key-value pairs	default_values · object
Source	enum	Required	Values: static env_var vault · default: static	source · enum
Override policy	enum	Required	Values: runtime_wins defaults_win · default: runtime_wins	override_policy · enum
Scope	enum	Optional	Values: node flow · default: node	scope · enum

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_DI_001	Invalid JSON	Default values must be valid JSON	json_parse_error: {reason}	Fix JSON syntax in editor
ERR_DI_002	Vault key not found	Secret key '{key}' was not found in the vault	vault_key_not_found: {key}	Add secret to vault or fix key name
ERR_DI_003	Env var not found	Environment variable '{var}' is not defined	env_var_not_found: {var}	Define env var in platform settings

Input / Output JSON

Input JSON	Output JSON { "run_id": "run_7b4e91d"
-------------------	---

```
{ "run_id": "run_7b4e91d", "payload":  
  { "source": "siem_inbound" } }
```

```
"defaults": { "me_api_base":  
  "https://me.internal/api/v3"  
  "me_api_key": ".....",  
  "critical_threshold": 1  
  "client_map": { "CLIENT_B":  
    "delta_inc" } } }
```

Logic Applied

Runtime wins over defaults. Secrets injected from vault at runtime. client_map is routing table for Router block. All defaults accessible as \$.defaults.* downstream.

10 — LISTENER NODE



Listener node

Shape: Hexagon
Colour: Coral / Red
Antenna dot: live connection status
Ports: 0 in | 1 out

Description

Persistent always-on event source. Spawns a new flow run for each matching event. Configured from Integrations → Listeners menu. Seven types: SIEM, Webhook, Email, File/Storage, Message Queue, Database/CDC, WebSocket/SSE.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Listener name	string	Required	Max 128 chars. Must reference a saved listener from Integrations menu	name · string
Listener type	enum	Required	Values: siem webhook email file message_queue database_cdc websocket	listener_type · enum
CONNECTION				
Connected account	string(uuid)	Required	Must reference an active saved listener instance	listener_id · uuid
Event filter	string	Optional	JSONPath predicate applied before spawning flow run	event_filter · string
DEDUP				
Dedup field	string	Optional	Valid JSONPath. E.g. \$.alert_id	dedup_field · string · JSONPath

Dedup window (s)	<code>integer</code>	Optional	Range: 0–3600 · default: 60	<code>dedup_window_s · integer</code>
OUTPUT				
Output key name	<code>string</code>	Optional	Max 64 chars · default: <code>listener_event</code>	<code>output_key · string</code>

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_001	Listener offline	The connected listener '{name}' is offline or disconnected	<code>listener_offline: listener_id:{id}</code>	Check listener config in Integrations
ERR_LI_002	Auth failure	Listener authentication failed for '{name}'	<code>listener_auth_error: listener_id:{id}</code>	Update credentials in Integrations
ERR_LI_003	Event filter error	Event filter expression could not be evaluated: {reason}	<code>event_filter_error: filter:{expr}</code>	Fix JSONPath filter expression
ERR_LI_004	Dedup suppressed	Event suppressed: duplicate '{dedup_field}' within dedup window	<code>event_dedup: field:{field} value:{val}</code>	Informational — no action needed

Input / Output JSON

Input JSON

```
{ "listener_type": "siem", "platform": "Splunk"
  "alert_id": "splk_882", "severity": "critical"
  "source_host": "WKST-042", "received_at": "2025-05-30T08:41:55Z" }
```

Output JSON

```
{ "flow_id": "flow_siem_master_v2",
  "run_id": "run_7b4e91d"
  "input": { "alert_id": "splk_882", "severity": "critical" }
  "started_at": "2025-05-30T08:41:55Z" }
```

Logic Applied

Listener spawns a new flow run per matching event. Payload lands at `$.input.*`. Dedup window suppresses duplicate `alert_id` within configured seconds.

11 — NOTIFICATION NODE



Notification node

Shape: Rectangle with bell icon
Colour: Amber
Email · Slack · Teams · SMS
Ports: 1 in | 1 out

Description

Sends an outbound message from within the flow. Configured from Integrations → Notifications menu. Phase 1: Email. Future: Slack, Teams, SMS, outbound webhook.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
ACCOUNT				
Notification account	string(uuid)	Required	Must reference an active saved account from Integrations → Notifications	notification_account_id · uuid
RECIPIENTS				
To	string	Required	Comma-separated emails. Each must be valid email format. Supports {{variable}}	to · string · email(s)
CC	string	Optional	Comma-separated emails. Each must be valid email format	cc · string · email(s)
CONTENT				
Subject	string	Required	Max 256 chars. Supports {{variable}}	subject · string
Body	string	Required	HTML or plain text. Max 50000 chars. Supports {{variable}}	body · string · html
Attachments	string[]	Optional	Max 5. Each must be a valid URL or flow artifact reference	attachments · string[]
DELIVERY				
On failure	enum	Optional	Values: retry skip fail_flow · default: retry	on_failure · enum
Retry count	integer	Optional	Range: 1–5 · default: 3	retry_count · integer

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_NO_001	Notification account not found	The notification account could not be found or is inactive	notification_account_not_found: {id}	Check Integrations → Notifications
ERR_NO_002	Invalid recipient	'{email}' is not a valid email address	invalid_email: {email}	Fix the To or CC field
ERR_NO_003	Delivery failed	Email delivery failed after {n} attempts: {reason}	delivery_failed: to:{email} reason:{reason}	Check SMTP config or recipient server
ERR_NO_004	Template render error	Could not render {{variable}} in subject or body: {reason}	template_render_error: field:{field}	Check variable references

Input / Output JSON

Input JSON

```
{ "account": "SOC Alerts Mailbox", "to": "analyst@company.com"
"subject": "[CRITICAL] Ransomware on WKST-042 - TK-2187"
"body": "SOP executed. Evidence: {{result.evidence_url}}" }
```

Output JSON

```
{ "notification_id": "notif_a3c2d1",
"status": "sent"
"delivered_at": "2025-05-30T09:38:01Z",
"recipient_count": 1 }
```

Logic Applied

Account credentials from Notifications menu. Subject and body template-rendered against flow context. Delivery status captured in notification_id for audit.

12 — LOOP NODE [NEW]



Loop node

Shape: Rounded rect + loop icon
Colour: Pink
Ports: 1 in · Body out · Done out
max_iterations safety cap built-in

Description
Dedicated iteration node supporting four loop types: for-each (iterate over a list), count (N times), while (check then loop), do-while (loop then check). Manages iteration state internally — no external counter wiring needed. Always has a max_iterations safety cap.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
Loop type	enum	Required	Values: for_each count while do_while · drives all visible fields below	loop_type · enum
FOR-EACH SETTINGS				
Source list	string	Conditional	Valid JSONPath to an array. Required when loop_type = for_each	source_list · string · JSONPath
Item variable	string	Conditional	Max 64 chars. Accessible as \$.loop.item · default: current_item	item_var · string
Index variable	string	Optional	Max 64 chars. Accessible as \$.loop.index · default: loop_index	index_var · string
Execution mode	enum	Conditional	Values: sequential parallel · Required for for_each and count	execution_mode · enum

Max concurrency	<code>integer</code>	Conditional	Range: 1–100 · Required when <code>execution_mode = parallel</code> · default: 10	<code>max_concurrency</code> · <code>integer</code>
Continue on item error	<code>boolean</code>	Optional	If true failed items logged and skipped · default: false	<code>continue_on_error</code> · <code>boolean</code>
COUNT SETTINGS				
Count	<code>integer string</code>	Conditional	Range: 1–10000. Supports <code>{{variable}}</code> · Required when <code>loop_type = count</code>	<code>count</code> · <code>integer</code> or <code>expression</code>
WHILE / DO-WHILE SETTINGS				
Condition type	<code>enum</code>	Conditional	Values: <code>expression ai_prompt</code> · Required for <code>while</code> and <code>do_while</code>	<code>condition_type</code> · <code>enum</code>
Condition expression	<code>string</code>	Conditional	Valid JSONPath predicate. E.g. <code>\$.status != 'resolved'</code>	<code>condition_expr</code> · <code>string</code>
AI prompt condition	<code>string</code>	Conditional	Min 10 chars. Max 500 chars. Must resolve to yes/no	<code>condition_prompt</code> · <code>string</code>
SAFETY CONTROLS — ALL LOOP TYPES				
Max iterations	<code>integer</code>	Required	Range: 1–100000 · default: 1000 · Hard ceiling enforced by engine	<code>max_iterations</code> · <code>integer</code>
On max reached	<code>enum</code>	Required	Values: <code>exit_done fail_flow notify_and_exit</code> · default: <code>notify_and_exit</code>	<code>on_max_action</code> · <code>enum</code>
Iteration timeout (s)	<code>integer</code>	Optional	Range: 1–3600 · default: 300	<code>iteration_timeout_s</code> · <code>integer</code>
Loop timeout (s)	<code>integer</code>	Optional	Range: 0–86400 · 0 = disabled · default: 3600	<code>loop_timeout_s</code> · <code>integer</code>
Watchdog interval (s)	<code>integer</code>	Optional	Range: 0–86400 · 0 = disabled. True wall-clock timer independent of iteration speed	<code>watchdog_interval_s</code> · <code>integer</code>
On loop timeout	<code>enum</code>	Conditional	Values: <code>exit_done fail_flow human_task</code> · Required when timeout set	<code>on_timeout_action</code> · <code>enum</code>
OUTPUT				
Collect results	<code>boolean</code>	Optional	Aggregate each iteration output into <code>\$.loop.results[]</code> · default: true	<code>collect_results</code> · <code>boolean</code>
Results key	<code>string</code>	Optional	Max 64 chars · default: <code>loop_results</code>	<code>results_key</code> · <code>string</code>
Collect errors	<code>boolean</code>	Optional	Failed iterations captured in <code>\$.loop.errors[]</code> · default: true	<code>collect_errors</code> · <code>boolean</code>

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LP_001	Source list not found	JSONPath ' <code>\$.{path}</code> ' did not resolve to an array	<code>source_list_not_found:</code> <code>path:{path}</code>	Fix JSONPath or check upstream data
ERR_LP_002	Max iterations reached	Loop reached the maximum iteration limit of <code>{max}</code>	<code>max_iterations_reached:</code> <code>limit:{max} loop:{name}</code>	Increase limit or fix exit condition

ERR_LP_003	Loop timeout exceeded	Loop ran for {elapsed}s exceeding the {timeout}s limit	loop_timeout: elapsed:{s} limit:{timeout}	Increase timeout or optimise body
ERR_LP_004	Iteration timeout	Iteration {index} timed out after {timeout_s}s	iteration_timeout: index:{i} limit:{s}	Optimise loop body or increase iter timeout
ERR_LP_005	Invalid while condition	The loop condition could not be evaluated: {reason}	condition_eval_error: expr:{expr}	Fix condition expression or prompt
ERR_LP_006	Watchdog fired	Loop watchdog triggered after {interval}s of wall-clock time	loop_watchdog: interval:{s} iterations:{n}	Human task launched to stop or continue

Input / Output JSON

Input JSON

```
{ "source_list": "$.sop.steps",
  "loop_type": "for_each"
  "item_var": "current_step",
  "execution_mode": "sequential"
  "max_iterations": 50 }
```

Output JSON

```
{ "loop_results": [
  { "step": "isolate_endpoint", "status":
    "success" }
  { "step": "run_forensics", "status":
    "success" } ]
  "loop_meta": { "iterations_run": 5,
    "exit_reason": "count_complete"
    "total_elapsed_s": 412 } }
```

Logic Applied

Body port fires into loop body each iteration with \$.loop.item, \$.loop.index, \$.loop.elapsed_s. Done port fires once when loop exits. exit_reason: count_complete | condition_false | break | max_iterations | timeout.

13 — WAIT NODE — TIMER [NEW]



Wait node — Timer

Shape: Horizontal hourglass

Colour: Orange

fixed · until_datetime · expression

Ports: 1 in · 1 out

Description

Pauses flow execution for a fixed duration, until a specific datetime, or until a datetime resolved from the flow context. Supports a reminder notification before the timer fires.

Configuration Fields — Field type · Validation · Dev key

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				

Name	<code>string</code>	Required	Max 128 chars	<code>name</code> · <code>string</code>
Wait type	<code>enum</code>	Required	Values: <code>timer</code> <code>process_wait</code> · must be <code>timer</code> for this sub-type	<code>wait_type</code> · <code>enum</code>
DURATION				
Duration type	<code>enum</code>	Conditional	Values: <code>fixed</code> <code>until_datetime</code> <code>expression</code> · Required when <code>wait_type</code> = <code>timer</code>	<code>duration_type</code> · <code>enum</code>
Duration value	<code>integer</code> <code>string</code>	Conditional	Range: 1–86400. Supports <code>{{variable}}</code> · Required when <code>duration_type</code> = <code>fixed</code>	<code>duration_value</code> · <code>integer</code> or <code>expression</code>
Duration unit	<code>enum</code>	Conditional	Values: <code>seconds</code> <code>minutes</code> <code>hours</code> <code>days</code> · Required when <code>duration_type</code> = <code>fixed</code>	<code>duration_unit</code> · <code>enum</code>
Wait until	<code>string(datetime)</code>	Conditional	ISO-8601 datetime. Supports <code>{{variable}}</code> · Required when <code>duration_type</code> = <code>until_datetime</code>	<code>wait_until</code> · <code>string</code> · ISO-8601
Wait until expression	<code>string</code>	Conditional	Valid JSONPath resolving to ISO-8601 datetime · Required when <code>duration_type</code> = <code>expression</code>	<code>wait_until_expr</code> · <code>string</code> · <code>JSONPath</code>
Timezone	<code>string</code>	Conditional	Valid IANA timezone · Required when <code>duration_type</code> = <code>until_datetime</code> · default: Asia/Kolkata	<code>timezone</code> · <code>string</code> · IANA
SAFETY				
Max wait (s)	<code>integer</code>	Optional	Range: 0–2592000 (30 days) · 0 = disabled	<code>max_wait_s</code> · <code>integer</code>
On timeout	<code>enum</code>	Conditional	Values: <code>continue</code> <code>fail_flow</code> <code>human_task</code> · Required when <code>max_wait</code> set · default: <code>human_task</code>	<code>on_timeout</code> · <code>enum</code>
REMINDER				
Reminder enabled	<code>boolean</code>	Optional	Fires a Notification node N seconds before timer ends · default: <code>false</code>	<code>reminder_enabled</code> · <code>boolean</code>
Reminder before (s)	<code>integer</code>	Conditional	Range: 1– <code>max_wait_s</code> · Required when <code>reminder_enabled</code> = <code>true</code>	<code>reminder_before_s</code> · <code>integer</code>

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_WT_001	Invalid datetime	The datetime '{value}' could not be parsed as ISO-8601	<code>datetime_parse_error:</code> <code>value:{val}</code>	Fix format: YYYY-MM-DDTHH:MM:SSZ
ERR_WT_002	Datetime in the past	The configured <code>wait_until</code> datetime has already passed	<code>datetime_already_past:</code> <code>value:{val}</code>	Update datetime or use a dynamic expression

ERR_WT_003	Max wait exceeded	Wait exceeded the maximum allowed duration of {max}s	max_wait_exceeded: limit:{s}	Check on_timeout action and increase if needed
ERR_WT_004	Expression not a datetime	The JSONPath expression '{path}' resolved to '{value}' which is not a valid datetime	wait_expr_not_datetime: path:{p} value:{v}	Ensure the field holds an ISO-8601 datetime

Input / Output JSON

Input JSON

```
{ "ticket_id": "TK-2187", "patch_applied": "KB5034441"
  "patch_applied_at": "2025-05-30T09:10:00Z" }
```

Output JSON

```
{ "ticket_id": "TK-2187", "patch_applied": "KB5034441"
  "wait_meta": { "wait_type": "timer"
    "wait_started_at": "2025-05-30T09:10:05Z"
    "wait_ended_at": "2025-05-30T09:40:05Z"
    "duration_s": 1800, "exit_reason": "timer_elapsed"
    "reminder_sent": true } }
```

Logic Applied

Flow suspended for exactly 30 min. All context passes through unchanged. wait_meta appended with timing and exit_reason: timer_elapsed | datetime_reached | timeout | resumed_by_human.

14 — WAIT NODE — PROCESS WAIT (ANY / ALL) [NEW]



Wait — Process (Any)

Shape: Hourglass + N inputs
 ANY = first branch unblocks flow
 Others cancelled automatically
 Ports: N in · 1 out



Wait — Process (All)

Shape: Hourglass + N inputs (thick)
 ALL = every branch must complete
 Merge: named_keys or merge_all
 Ports: N in · 1 out

Description

Synchronisation gate for parallel branches. ANY mode fires when the first of N branches arrives (others cancelled). ALL mode fires only when every branch has arrived and merges all payloads. Number of input ports equals expected_branches.

Configuration Fields

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
Wait type	enum	Required	Values: timer process_wait · must be process_wait	wait_type · enum
PROCESS WAIT SETTINGS				
Wait mode	enum	Required	Values: any all · drives port count and merge logic	wait_mode · enum
Expected branches	integer	Required	Range: 2–20 · sets number of input ports	expected_branches · integer
Branch labels	string[]	Optional	Max 64 chars each. Count must equal expected_branches	branch_labels · string[]
Branch timeout (s)	integer	Optional	Range: 0–86400 · 0 = wait forever · default: 3600	branch_timeout_s · integer
On branch timeout	enum	Conditional	Values: continue_with_arrived fail_flow human_task · Required when timeout set	on_branch_timeout · enum
MERGE STRATEGY				
Merge strategy	enum	Optional	Values: merge_all first_only named_keys · default: merge_all	merge_strategy · enum
Branch output keys	string[]	Conditional	Max 64 chars each. Count must equal expected_branches · Required when merge_strategy = named_keys	branch_output_keys · string[]
Include branch metadata	boolean	Optional	Appends \$.wait_meta with timing and branch arrival info · default: true	include_meta · boolean

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_PW_001	Branch timeout	Not all branches arrived within {timeout}s. Arrived: {n}/{total}	branch_timeout: arrived:{n} expected:{total}	Human task lists pending branches
ERR_PW_002	Unexpected branch	An unregistered branch attempted to connect	unexpected_branch: branch_id:{id}	Adjust expected_branches count
ERR_PW_003	Merge conflict	Two branches returned the same key with different values in merge_all mode	merge_conflict: key:{key} values:{v1},{v2}	Use named_keys merge strategy
ERR_PW_004	Wrong branch count	Number of branch_output_keys must equal expected branches	branch_key_count_mismatch: keys:{n} branches:{m}	Align branch_output_keys count

Input JSON	Output JSON
------------	-------------

```
{ (ALL mode) branch 1: { "client_id":  
"CLIENT_B", "sop_success": true,  
"patch_kb": "KB5034441" }  
  branch 2: { "client_id": "CLIENT_C",  
"sop_success": true, "legal_hold_id": "LH-  
2025-441" } }
```

```
{ "delta": { "client_id": "CLIENT_B",  
"sop_success": true, "patch_kb":  
"KB5034441" }  
  "gamma": { "client_id": "CLIENT_C",  
"sop_success": true, "legal_hold_id": "LH-  
2025-441" }  
  "wait_meta": { "wait_mode": "all",  
"arrived_branches": ["delta","gamma"]  
  "all_succeeded": true, "exit_reason":  
"all_branches_arrived"  
  "total_wait_s": 1652 } }
```

Logic Applied

ALL mode: holds until all N branches arrive. named_keys merge stores each branch under configured key. all_succeeded computed from sop_success across branches. ANY mode: first branch fires, others cancelled. first_only merge passes only the winning branch payload downstream.

Section 3: Flow-Level Triggers — Scheduler & Webhook

Scheduler and Webhook are per-flow triggers configured inside the flow settings panel (Flow editor → Settings → Triggers tab). They are NOT in the shared Integrations menu because they belong to exactly one flow each.

3.1 Scheduler Trigger



Scheduler trigger

Shape: Dashed circle
Colour: Gray
Cron: interval · fixed time
Ports: 0 in | 1 out (per-flow)

Field label	Field type	Required	Validation rule	Dev key · notes
ENABLE				
Enable scheduler	boolean	Required	—	scheduler_enabled · boolean · default: false
SCHEDULE				
Schedule type	enum	Required	Values: interval cron fixed_time	schedule_type · enum
Interval value	integer string	Conditional	Range: 1–10000. Supports {{variable}} · Required when schedule_type = interval	interval_value · integer
Interval unit	enum	Conditional	Values: minutes hours days · Required when schedule_type = interval	interval_unit · enum
Cron expression	string	Conditional	Valid 5-field cron syntax. Live preview shown · Required when schedule_type = cron	cron_expression · string
Run at time	string	Conditional	Format HH:MM (24h) · Required when schedule_type = fixed_time	run_at_time · string · HH:MM
Run on days	string[]	Conditional	Values from Mon Tue Wed Thu Fri Sat Sun · Required when schedule_type = fixed_time	run_on_days · string[]
Timezone	string	Required	Valid IANA timezone · default: Asia/Kolkata	timezone · string · IANA
BEHAVIOUR				
Start date	string	Optional	ISO-8601 date · default: immediately	start_date · string · date
End date	string	Optional	ISO-8601 date · leave blank = run forever	end_date · string · date

Overlap policy	enum	Optional	Values: skip queue run_anyway · default: skip	overlap_policy · enum
Catch-up missed runs	boolean	Optional	Fire missed runs after restart · default: false	catchup · boolean
INPUT				
Static input payload	object	Optional	Valid JSON. Injected as \$.input on each scheduled run	static_input · object

Scheduler Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_SC_001	Invalid cron	Cron expression is not valid: {expr}	cron_parse_error: {expr}	Fix using cron preview helper
ERR_SC_002	Overlapping run skipped	A run was skipped because a previous run is still active	overlap_skipped: active_run:{id}	Adjust overlap_policy or optimise flow
ERR_SC_003	Missed run catch-up	Catch-up firing {n} missed run(s) after restart	catchup_runs: count:{n}	Informational — review catchup setting

3.2 Webhook Trigger



Field label	Field type	Required	Validation rule	Dev key · notes
ENABLE				
Enable webhook	boolean	Required	—	webhook_enabled · boolean · default: false
ENDPOINT				
Webhook URL	string	Optional	Read-only. Auto-generated. https://hooks.flowAutomate.io/flows/{id}/trigger	webhook_url · string · computed
HTTP method	enum	Required	Values: POST GET PUT PATCH · default: POST	webhook_method · enum
RESPONSE				

Response mode	enum	Required	Values: async_202 sync_with_result · default: async_202	response_mode · enum
Sync timeout (s)	integer	Conditional	Range: 1–30 · Required when response_mode = sync_with_result	sync_timeout_s · integer
Custom ack body	object	Optional	Valid JSON. Supports {{run_id}} and {{flow_id}} tokens	custom_ack_body · object
AUTH				
Auth method	enum	Required	Values: none api_key hmac_sha256 jwt_bearer · default: api_key	auth_method · enum
HMAC header	string	Conditional	Max 128 chars · Required when auth_method = hmac_sha256 · default: X-Flow-Signature	hmac_header · string
JWT audience	string	Conditional	Max 256 chars · Required when auth_method = jwt_bearer	jwt_audience · string
JWT JWKS URL	string	Conditional	Must be valid HTTPS URL · Required when auth_method = jwt_bearer	jwt_jwks_url · string · url
VALIDATION				
Body schema	object	Optional	Valid JSON Schema draft-07. Invalid bodies rejected with 422	body_schema · object
Max body size (KB)	integer	Optional	Range: 1–102400 · default: 1024	max_body_kb · integer
RATE LIMITING				
Max requests / min	integer	Optional	Range: 0–10000 · 0 = unlimited · default: 60	rate_limit_rpm · integer
IP allowlist	string	Optional	CIDR format. Comma-separated. Blank = allow all	ip_allowlist · string · CIDR
API SCHEMA				
Input parameters	object	Optional	JSON Schema. Fields land as \$.input.* in flow context. Auto-generates API docs	input_schema · object
Output parameters	object	Optional	JSONPath mapping. Extracts fields from final context for API response	output_schema · object

Webhook Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_WH_001	HMAC mismatch	Webhook signature verification failed	hmac_mismatch: expected:{hash}	Verify caller HMAC signing secret
ERR_WH_002	Body schema invalid	Request body does not match configured schema: {field} {reason}	schema_validation_error: {field}	Fix caller payload or relax schema
ERR_WH_003	Body too large	Request body ({size}KB) exceeds max_body_kb limit ({limit}KB)	body_too_large: size:{n}kb limit:{m}kb	Increase limit or reduce payload
ERR_WH_004	Rate limit exceeded	Too many requests. Retry after {reset_ms}ms	rate_limit_exceeded: rpm:{limit}	Increase limit or throttle caller

ERR_WH_005	IP not allowlisted	Request from {ip} is not in the configured allowlist	ip_not_allowed: ip:{ip}	Add IP to allowlist or remove restriction
ERR_WH_006	Sync timeout	Flow did not complete within the sync timeout of {timeout}s	sync_timeout: after:{s}s run_id:{id}	Increase sync_timeout or switch to async

Section 4: Integrations Menu — Phase 1 Configuration

Shared platform-level configurations. Phase 1: 7 Listener types, 1 Connector (HTTP/REST), 1 Notification (Email Account). Each listener/connector/notification is configured once and referenced by multiple flows.

4.1 SIEM — Listener

Live alert feed from Splunk, QRadar, Sentinel, Elastic, Chronicle, Sumo Logic, Cortex XSOAR.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars · unique per tenant	name · string
Description	string	Optional	Max 512 chars	description · string
PLATFORM				
SIEM platform	enum	Required	Values: splunk qradar sentinel elastic chronicle sumo_logic cortex custom	platform · enum
Connection mode	enum	Required	Values: rest_api syslog ceff leef webhook_push	connection_mode · enum
REST API				
Base URL	string	Conditional	Must start with https://. Max 2048 chars · Required when connection_mode = rest_api	base_url · string · url
API token	string(secret)	Conditional	Min 8 chars · Required when connection_mode = rest_api	api_token · vault_ref
Verify TLS	boolean	Optional	—	verify_tls · boolean · default: true
Poll interval (s)	integer	Conditional	Range: 10–3600 · Required when connection_mode = rest_api · default: 30	poll_intervals · integer
SYSLOG				
Listen port	integer	Conditional	Range: 1–65535 · Required when connection_mode = syslog	listen_port · integer · default: 514
Protocol	enum	Conditional	Values: udp tcp tls_tcp · Required when connection_mode = syslog	protocol · enum
FILTERING				
Min severity	enum	Optional	Values: low medium high critical · default: medium	min_severity · enum
Rule name pattern	string	Optional	Regex. Max 256 chars	rule_name_pattern · string · regex

Dedup field	<code>string</code>	Optional	Valid JSONPath. E.g. \$.alert_id	dedup_field · string · JSONPath
Dedup window (s)	<code>integer</code>	Optional	Range: 0–3600 · default: 60	dedup_window_s · integer
OUTPUT				
Output key name	<code>string</code>	Optional	Max 64 chars · default: siem_event	output_key · string
Status	<code>enum</code>	Required	Values: active paused · default: active	status · enum

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	auth_error: {reason}	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	connect_error: host:{h} reason:{r}	Check host, port, TLS settings
ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	disconnect: listener:{id}	Automatic reconnect — check logs

4.2 Webhook — Listener

Inbound HTTP from any external system. Auto-generates a unique listener endpoint URL.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	<code>string</code>	Required	Max 128 chars	name · string
ENDPOINT				
URL slug	<code>string</code>	Required	Lowercase alphanumeric and hyphens only. Max 64 chars. Auto-suggested from name	slug · string · slug
HTTP methods	<code>string[]</code>	Required	Min 1. Values: POST GET PUT PATCH · default: [POST]	allowed_methods · string[]
Response mode	<code>enum</code>	Required	Values: async_202 sync_with_result · default: async_202	response_mode · enum
AUTH				
Auth method	<code>enum</code>	Required	Values: none api_key hmac_sha256 mtls jwt_bearer · default: api_key	auth_method · enum
HMAC secret	<code>string(secret)</code>	Conditional	Min 16 chars · Required when auth_method = hmac_sha256	hmac_secret · vault_ref
VALIDATION				

Body schema	<code>object</code>	Optional	Valid JSON Schema draft-07	<code>body_schema</code> · <code>object</code>
Max body size (KB)	<code>integer</code>	Optional	Range: 1–102400 · default: 1024	<code>max_body_kb</code> · <code>integer</code>
OUTPUT				
Output key name	<code>string</code>	Optional	Max 64 chars · default: <code>webhook_payload</code>	<code>output_key</code> · <code>string</code>
Status	<code>enum</code>	Required	Values: <code>active</code> <code>paused</code> · default: <code>active</code>	<code>status</code> · <code>enum</code>

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	<code>auth_error: {reason}</code>	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	<code>connect_error: host:{h} reason:{r}</code>	Check host, port, TLS settings
ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	<code>disconnect: listener:{id}</code>	Automatic reconnect — check logs
ERR_LI_SLUG	Slug already taken	URL slug '{slug}' is already in use	<code>slug_conflict: {slug}</code>	Choose a different slug

4.3 Email — Listener

Trigger flows from inbound email. Monitors a mailbox via IMAP IDLE (push) or polling.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	<code>string</code>	Required	Max 128 chars	<code>name</code> · <code>string</code>
MAILBOX				
Protocol	<code>enum</code>	Required	Values: <code>imap</code> <code>exchange_ews</code> <code>gmail_api</code>	<code>protocol</code> · <code>enum</code>
Host	<code>string</code>	Conditional	Max 256 chars · Required when <code>protocol = imap</code>	<code>host</code> · <code>string</code>
Port	<code>integer</code>	Conditional	Range: 1–65535 · default: 993 · Required when <code>protocol = imap</code>	<code>port</code> · <code>integer</code>
TLS	<code>boolean</code>	Optional	—	<code>use_tls</code> · <code>boolean</code> · default: <code>true</code>
Username	<code>string</code>	Required	Valid email address format. Max 256 chars	<code>username</code> · <code>string</code> · <code>email</code>
Password / token	<code>string(secret)</code>	Required	Min 8 chars	<code>password</code> · <code>vault_ref</code>
Mailbox folder	<code>string</code>	Optional	Max 256 chars · default: <code>INBOX</code>	<code>folder</code> · <code>string</code>

Fetch mode	<code>enum</code>	Required	Values: <code>imap_idle poll</code> · default: <code>imap_idle</code>	<code>fetch_mode</code> · <code>enum</code>
Poll interval (s)	<code>integer</code>	Conditional	Range: 30–3600 · Required when <code>fetch_mode = poll</code> · default: 60	<code>poll_interval_s</code> · <code>integer</code>
FILTERING				
From pattern	<code>string</code>	Optional	Regex. Max 256 chars	<code>from_pattern</code> · <code>string</code> · <code>regex</code>
Subject pattern	<code>string</code>	Optional	Regex. Max 256 chars	<code>subject_pattern</code> · <code>string</code> · <code>regex</code>
Require attachment	<code>boolean</code>	Optional	—	<code>require_attachment</code> · <code>boolean</code> · default: <code>false</code>
Allowed attachment types	<code>string[]</code>	Optional	Values: <code>pdf csv json xml txt</code>	<code>attachment_types</code> · <code>string[]</code>
POST-PROCESSING				
Parse attachment	<code>boolean</code>	Optional	—	<code>parse_attachment</code> · <code>boolean</code> · default: <code>false</code>
Mark as read	<code>boolean</code>	Optional	—	<code>mark_read</code> · <code>boolean</code> · default: <code>true</code>
OUTPUT				
Output key name	<code>string</code>	Optional	Max 64 chars · default: <code>email_event</code>	<code>output_key</code> · <code>string</code>
Status	<code>enum</code>	Required	Values: <code>active paused</code>	<code>status</code> · <code>enum</code>

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	<code>auth_error: {reason}</code>	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	<code>connect_error: host:{h} reason:{r}</code>	Check host, port, TLS settings
ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	<code>disconnect: listener:{id}</code>	Automatic reconnect — check logs

4.4 File / Storage — Listener

Watch a bucket, folder, or share for new or changed files.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	<code>string</code>	Required	Max 128 chars	<code>name</code> · <code>string</code>
STORAGE				

Storage type	enum	Required	Values: s3 azure_blob gcs sftp smb local	storage_type · enum
Bucket / container	string	Conditional	Max 256 chars · Required for cloud types	bucket · string
Region	string	Conditional	AWS region format e.g. ap-south-1 · Required for S3	region · string
SFTP host	string	Conditional	Max 256 chars · Required when storage_type = sftp	sftp_host · string
SFTP port	integer	Conditional	Range: 1–65535 · default: 22 · Required when storage_type = sftp	sftp_port · integer
Access key ID	string	Conditional	Max 128 chars · Required for S3	access_key_id · string
Secret access key	string(secret)	Conditional	Min 16 chars · Required for S3	secret_access_key · vault_ref
Username	string	Conditional	Max 256 chars · Required for SFTP/SMB	username · string
Password / SSH key	string(secret)	Conditional	Min 8 chars · Required for SFTP/SMB	password · vault_ref
WATCH RULES				
Watch path	string	Required	Glob-supported. Max 512 chars	watch_path · string · glob
Trigger on	string[]	Required	Values: created modified deleted · Min 1	trigger_on · string[]
File extensions	string[]	Optional	E.g. [".json",".csv"]. Each max 10 chars	file_extensions · string[]
Poll interval (s)	integer	Required	Range: 10–3600 · default: 30	poll_interval_s · integer
CONTENT PARSING				
Parse content	boolean	Optional	—	parse_content · boolean · default: false
Content format	enum	Conditional	Values: json csv xml text binary_b64 · Required when parse_content = true	content_format · enum
OUTPUT				
Output key name	string	Optional	Max 64 chars · default: file_event	output_key · string
Status	enum	Required	Values: active paused	status · enum

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	auth_error: {reason}	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	connect_error: host:{h} reason:{r}	Check host, port, TLS settings

ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	disconnect: listener:{id}	Automatic reconnect — check logs
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4.5 Message Queue — Listener

Consume from Kafka, AWS SQS, RabbitMQ, or Azure Service Bus.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
BROKER				
Broker type	enum	Required	Values: kafka sqs rabbitmq azure_service_bus google_pub_sub	broker_type · enum
KAFKA				
Bootstrap servers	string	Conditional	Comma-separated host:port pairs · Required when broker_type = kafka	bootstrap_servers · string
Topic	string	Conditional	Max 256 chars · Required when broker_type = kafka	topic · string
Consumer group ID	string	Conditional	Max 256 chars · Required when broker_type = kafka	consumer_group_id · string
SASL mechanism	enum	Conditional	Values: none plain scram_sha256 scram_sha512	sasl_mechanism · enum
SASL username	string	Conditional	Max 256 chars · Required when SASL enabled	sasl_username · string
SASL password	string(secret)	Conditional	Min 8 chars · Required when SASL enabled	sasl_password · vault_ref
AWS SQS				
Queue URL	string	Conditional	Must be valid SQS URL · Required when broker_type = sqs	queue_url · string · url
AWS region	string	Conditional	AWS region format · Required when broker_type = sqs	aws_region · string
AWS access key ID	string	Conditional	Max 128 chars · Required when broker_type = sqs	aws_access_key_id · string
AWS secret key	string(secret)	Conditional	Min 16 chars · Required when broker_type = sqs	aws_secret_key · vault_ref
RABBITMQ				
AMQP URL	string	Conditional	amqps:// scheme required · Required when broker_type = rabbitmq	amqp_url · string · url
Queue name	string	Conditional	Max 256 chars · Required when broker_type = rabbitmq	queue_name · string

CONSUMPTION				
Batch size	integer	Optional	Range: 1–100 · default: 1	batch_size · integer
On failure	enum	Required	Values: requeue dlq discard · default: requeue	on_failure · enum
Output key	string	Optional	Max 64 chars · default: mq_message	output_key · string
Status	enum	Required	Values: active paused	status · enum

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	auth_error: {reason}	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	connect_error: host:{h} reason:{r}	Check host, port, TLS settings
ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	disconnect: listener:{id}	Automatic reconnect — check logs
ERR_LI_LAG	Consumer lag too high	Message queue consumer lag exceeds {threshold}	consumer_lag: topic:{t} lag:{n}	Scale consumers or reduce flow duration

4.6 Database / CDC — Listener

Change Data Capture on table insert, update, or delete. Supports Debezium, polling, and DB triggers.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Name	string	Required	Max 128 chars	name · string
DATABASE				
DB type	enum	Required	Values: postgresql mysql mssql oracle	db_type · enum
Host	string	Required	Max 256 chars	host · string
Port	integer	Required	Range: 1–65535 · defaults: 5432/3306/1433	port · integer
Database name	string	Required	Max 128 chars	database · string
Username	string	Required	Max 128 chars	username · string
Password	string(secret)	Required	Min 8 chars	password · vault_ref
SSL mode	enum	Optional	Values: disable require verify-ca verify-full · default: require	ssl_mode · enum
CDC METHOD				

CDC method	enum	Required	Values: debezium polling db_trigger	cdc_method · enum
Replication slot	string	Conditional	Max 64 chars · Required when cdc_method = debezium and db_type = postgresql	replication_slot · string
Poll interval (s)	integer	Conditional	Range: 5–3600 · Required when cdc_method = polling · default: 30	poll_interval_s · integer
Watermark column	string	Conditional	Max 128 chars · Required when cdc_method = polling	watermark_column · string
WATCH RULES				
Table name	string	Required	Max 128 chars	table · string
Operations	string[]	Required	Values: INSERT UPDATE DELETE · Min 1 · default: [INSERT]	operations · string[]
Row filter	string	Optional	SQL WHERE clause. Max 512 chars	row_filter · string · SQL
OUTPUT				
Output key	string	Optional	Max 64 chars · default: db_event	output_key · string
Include before image	boolean	Optional	—	include_before · boolean · default: false
Status	enum	Required	Values: active paused	status · enum

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	auth_error: {reason}	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	connect_error: host:{h} reason:{r}	Check host, port, TLS settings
ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	disconnect: listener:{id}	Automatic reconnect — check logs
ERR_LI_CDC	Replication slot full	WAL replication slot is full. Data loss risk	slot_full: slot:{name}	Increase WAL retention or consumer speed

4.7 WebSocket / SSE — Listener

Persistent real-time event stream. Supports wss:// (WebSocket) and https:// (Server-Sent Events). Auto-reconnects with exponential backoff.

Configuration

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				

Name	<code>string</code>	Required	Max 128 chars	<code>name · string</code>
CONNECTION				
Protocol	<code>enum</code>	Required	Values: websocket sse	<code>protocol · enum</code>
URL	<code>string</code>	Required	Must start with wss:// (WebSocket) or https:// (SSE). Max 2048 chars	<code>url · string · url</code>
Custom headers	<code>object</code>	Optional	String→string map. Max 20 headers	<code>headers · object</code>
AUTH				
Auth type	<code>enum</code>	Optional	Values: none bearer_token api_key_header basic · default: none	<code>auth_type · enum</code>
Auth value	<code>string(secret)</code>	Conditional	Min 8 chars · Required when <code>auth_type != none</code>	<code>auth_value · vault_ref</code>
Auth header name	<code>string</code>	Conditional	Max 128 chars · Required when <code>auth_type = api_key_header</code> · default: X-API-Key	<code>auth_header · string</code>
RECONNECT				
Auto reconnect	<code>boolean</code>	Optional	—	<code>auto_reconnect · boolean · default: true</code>
Initial backoff (ms)	<code>integer</code>	Optional	Range: 100–60000 · default: 1000	<code>reconnect_initial_ms · integer</code>
Max backoff (ms)	<code>integer</code>	Optional	Range: 1000–300000 · default: 30000	<code>reconnect_max_ms · integer</code>
Ping interval (s)	<code>integer</code>	Optional	Range: 5–300 · default: 30 · WebSocket only	<code>ping_interval_s · integer</code>
EVENT FILTERING				
Event type filter	<code>string</code>	Optional	Regex or exact match. Max 256 chars	<code>event_type_filter · string</code>
JSON filter expression	<code>string</code>	Optional	JSONPath predicate. E.g. <code>\$.severity == 'critical'</code>	<code>filter_expression · string · JSONPath</code>
OUTPUT				
Output key	<code>string</code>	Optional	Max 64 chars · default: <code>stream_event</code>	<code>output_key · string</code>
Status	<code>enum</code>	Required	Values: active paused	<code>status · enum</code>

Error Messages

Error code	Trigger	UI message	Dev message	Recovery action
ERR_LI_AUTH	Authentication failure	Could not authenticate to the external system	<code>auth_error: {reason}</code>	Update credentials
ERR_LI_CONN	Connection failed	Could not connect to the server at {host}	<code>connect_error: host:{h} reason:{r}</code>	Check host, port, TLS settings
ERR_LI_DISC	Disconnected	Listener lost connection. Reconnecting...	<code>disconnect: listener:{id}</code>	Automatic reconnect — check logs

ERR_LI_WS	Max reconnect reached	WebSocket reconnection failed after {n} attempts	max_reconnect: attempts:{n}	Check server availability and auth
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4.8 Connector — HTTP / REST

See Section 2, Node 08 (Connector / API Call Node) for the full HTTP/REST and MCP tool configuration fields, validations, and error messages.

4.9 Notification — Email Account

Field label	Field type	Required	Validation rule	Dev key · notes
IDENTITY				
Account name	string	Required	Max 128 chars · unique per tenant	name · string
Display name (From)	string	Required	Max 128 chars	from_name · string
From address	string	Required	Valid email format. Max 256 chars	from_address · string · email
Reply-to	string	Optional	Valid email format. Max 256 chars	reply_to · string · email
PROVIDER				
Provider	enum	Required	Values: smtp sendgrid aws_ses mailgun postmark	provider · enum
SMTP host	string	Conditional	Max 256 chars · Required when provider = smtp	smtp_host · string
SMTP port	integer	Conditional	Range: 1–65535 · default: 587 · Required when provider = smtp	smtp_port · integer
Encryption	enum	Conditional	Values: none starttls ssl_tls · Required when provider = smtp · default: starttls	smtp_encryption · enum
SMTP username	string	Conditional	Max 256 chars · Required when provider = smtp	smtp_username · string
SMTP password	string(secret)	Conditional	Min 8 chars · Required when provider = smtp	smtp_password · vault_ref
API key	string(secret)	Conditional	Min 8 chars · Required for SendGrid SES Mailgun Postmark	api_key · vault_ref
AWS region	string	Conditional	AWS region format · Required when provider = aws_ses	aws_region · string
DEFAULTS				
Default recipients	string	Optional	Comma-separated emails. Each valid email format	default_to · string
Default subject	string	Optional	Max 256 chars. Supports {{variable}}	default_subject · string

Default body	<code>string</code>	Optional	HTML. Max 50000 chars. Supports {{variable}}	<code>default_body · string · html</code>
DELIVERY				
Max recipients	<code>integer</code>	Optional	Range: 1–200 · default: 50	<code>max_recipients · integer</code>
Retry count	<code>integer</code>	Optional	Range: 1–10 · default: 3	<code>retry_count · integer</code>

Section 5: Global Error Codes & Platform Error Handling

These error codes apply platform-wide across all node executions, independent of node type.

Error code	Trigger	UI message	Dev message	Recovery action
ERR_FLOW_001	Flow not found	This flow could not be found. It may have been deleted	flow_not_found: {flow_id}	Check flow ID in URL
ERR_FLOW_002	Flow validation failed	The flow has {n} error(s) that must be resolved before publishing	flow_invalid: errors:{{list}}	Fix all canvas validation errors
ERR_FLOW_003	Context payload too large	The flow context exceeded the 10 MB limit at node '{name}'	context_too_large: size:{mb}mb node:{name}	Reduce data passed between nodes
ERR_FLOW_004	Run timeout	The flow run exceeded the maximum execution time	run_timeout: run_id:{id} elapsed:{s}s	Increase timeout or split into sub-flows
ERR_FLOW_005	Vault secret unavailable	A required vault secret could not be retrieved	vault_unavailable: key:{key}	Check vault health and secret name
ERR_FLOW_006	Circular dependency	The flow contains a circular dependency: {path}	circular_dependency: {node_a} → {node_b}	Remove the cycle
ERR_FLOW_007	Max concurrent runs	The flow has reached its maximum concurrent run limit of {limit}	max_concurrent_runs: limit:{n}	Increase limit or queue runs
ERR_FLOW_008	Template render error	Could not render {{variable}} expression: {expr}	template_error: expr:{e} reason:{r}	Check variable name and context path
ERR_FLOW_009	JSONPath error	JSONPath expression could not be evaluated: {expr}	jsonpath_error: path:{p} reason:{r}	Fix JSONPath syntax
ERR_FLOW_010	Node execution error	An unexpected error occurred in node '{name}'	node_error: node:{name} type:{type} reason:{r}	Check node config and contact support

Section 6: Master Orchestration Flow — Node-by-Node JSON

Running example: ticket TK-2187, Delta Inc, ransomware WannaCry on WKST-042.

NODE 1 — START | INGESTION

Input JSON

```
{ "trigger": "webhook", "source": "siem_inbound", "timestamp": "2025-05-30T08:42:11Z", "metadata": { "hmac_verified": true } }
```

Output JSON

```
{ "flow_id": "flow_siem_master_v2", "run_id": "run_7b4e91d", "triggered_by": "webhook", "started_at": "2025-05-30T08:42:11Z" }
```

Logic Applied

HMAC verified. run_id generated (UUID v4). Failure exits immediately to failure End.

NODE 2 — DEFAULT INPUTS | INGESTION

Input JSON

```
{ "run_id": "run_7b4e91d", "payload": { "source": "siem_inbound" } }
```

Output JSON

```
{ "run_id": "run_7b4e91d", "defaults": { "me_api_base": "https://me.internal/api/v3", "me_api_key": ".....", "critical_threshold": 1, "client_map": { "CLIENT_B": "delta_inc" } } }
```

Logic Applied

Runtime wins over defaults. Vault secrets injected at runtime. client_map drives Router.

NODE 3 — API CALL: MANAGEENGINE FETCH | INGESTION

Input JSON

```
{ "method": "GET", "url": "https://me.internal/api/v3/tickets/new", "query_params": { "status": "open", "limit": 1 } }
```

Output JSON

```
{ "status_code": 200, "response": { "ticket_id": "TK-2187", "client_id": "CLIENT_B", "severity": "critical", "sop": { "sop_id": "SOP-EDR-001", "steps": ["isolate_endpoint", "run_forensics", "remediate"] } } }
```

Logic Applied

Full ticket returned with SOP steps. Retry 3x on error. Persistent failure → failure End + Slack.

NODE 4 — AGENT: TICKET PARSER | ENRICHMENT

Input JSON

```
{ "goal": "Parse SIEM ticket, extract structured fields"
  "context": { "ticket_id": "TK-2187",
    "severity": "critical" }, "max_steps": 1 }
```

Output JSON

```
{ "client_name": "delta_inc",
  "severity_rank": 1
  "incident": { "type": "malware",
    "subtype": "ransomware" }
  "requires_human_approval": true,
  "confidence": 0.97 }
```

Logic Applied

Classifies ransomware. Sets requires_human_approval: true. Confidence 0.97 > 0.85 — no escalation.

NODE 5 — CONDITIONAL: SEVERITY CRITICAL? | APPROVAL GATE

Input JSON

```
{ "severity": "critical", "severity_rank": 1 }
```

Output JSON

```
{ "severity": "critical", "severity_rank": 1
  "_branch": "true", "_evaluated_as": "1 <= 1 = true" }
```

Logic Applied

1 <= 1 = true. True → Human task. False → Router directly.

NODE 6 — HUMAN TASK: ANALYST APPROVAL | APPROVAL GATE

Input JSON

```
{ "assignee": "group:SOC_L1", "task_title": "Approve SOP for TK-2187"
  "due_in_minutes": 15, "decision_options": ["approve","reject"] }
```

Output JSON

```
{ "decision": "approve", "completed_by": "analyst_raj.kumar"
  "completed_at": "2025-05-30T08:49:03Z", "task_duration_s": 412 }
```

Logic Applied

Tokenised email sent to SOC_L1. SLA 15 min. Analyst identity merged into context.

NODE 7 — ROUTER: CLIENT ID | ROUTING

Input JSON

```
{ "client_id": "CLIENT_B", "decision": "approve" }
```

Output JSON

```
{ "client_id": "CLIENT_B", "_route": "CLIENT_B"
  "_route_label": "delta_inc_sop_flow", "_rule_index": 1 }
```

Logic Applied

Rule 1: CLIENT_B matches. First_match mode. delta_inc_sop_flow used as child flow ID.

NODE 8 — PROCESS CALL: DELTA_INC_SOP_FLOW | SOP EXECUTION

Input JSON

```
{ "sub_flow_id": "delta_inc_sop_flow",  
  "timeout_s": 300  
  "input_mapping": { "ticket_id": "TK-2187",  
    "hostname": "WKST-042" } }
```

Output JSON

```
{ "sub_run_id": "run_delta_c3f2a1",  
  "status": "completed"  
  "result": { "sop_success": true,  
    "patch_kb": "KB5034441"  
    "vlan_quarantined": "VLAN-14" } }
```

Logic Applied

Child runs synchronously (300s max). Secrets never passed. Returns sop_success: true.

NODE 9 — API CALL: UPDATE MANAGEENGINE | CLOSURE

Input JSON

```
{ "method": "PATCH", "url":  
  "https://me.internal/api/v3/tickets/TK-  
  2187"  
  "body": { "status": "resolved",  
    "evidence_url":  
    "https://evidence.internal/..." } }
```

Output JSON

```
{ "status_code": 200  
  "response": { "ticket_id": "TK-2187",  
    "status": "resolved"  
    "audit_entry_id": "audit_4d8c1a" } }
```

Logic Applied

PATCH with full resolution payload. Non-200 → retry once; second failure → partial_success.

NODE 10 — CONDITIONAL: SOP SUCCEEDED? | CLOSURE

Input JSON

```
{ "result": { "sop_success": true },  
  "me_update_status_code": 200 }
```

Output JSON

```
{ "_branch": "true"  
  "_evaluated_as": "true AND true = true" }
```

Logic Applied

Compound: sop_success AND status 200. Both true → success End. Either false → L2 escalation.

NODE 11 — END | CLOSURE

Input JSON

```
{ "ticket_id": "TK-2187", "result":  
  { "sop_success": true } }
```

Output JSON

```
{ "flow_id": "flow_siem_master_v2",  
  "run_id": "run_7b4e91d"  
  "status": "success", "duration_ms":  
  3331000  
  "audit_trail": { "me_audit_entry_id":  
    "audit_4d8c1a" } }
```

Logic Applied

Final run record assembled. audit_trail preserves analyst identity for compliance reporting.

Section 7: Client-Specific SOP Flows

Each sub-flow is called by the master flow Router.

7.1 Client A — Bravo Corp: Brute-force Login

TK-1042 — BRAVO CORP — SEVERITY: HIGH

SOP steps

- 1. Default inputs — Bravo Corp AD credentials and firewall API key
- 2. API call (MCP tool: iam_manager · disable_user) — disable j.smith in Active Directory
- 3. Agent — geo anomaly check: compare login IP to user baseline
- 4. Conditional — Foreign IP? Yes: API call (MCP tool: firewall_manager · block_ip)
- 5. Human task — notify user · re-enable or permanent ban?
- 6. Conditional — Re-enable? Yes: API call (MCP tool: iam_manager · enable_user)
- 7. End

Key decision-point JSON

Decision point	Sample JSON
Agent geo output	{ "login_ip": "195.44.21.3", "login_country": "RU", "geo_anomaly": true }
Firewall block output	{ "rule_id": "fw_rule_2291", "ip_blocked": "195.44.21.3", "status": "active" }
Human task decision	{ "decision": "ban_permanent", "completed_by": "analyst_priya.sharma" }

7.2 Client B — Delta Inc: Ransomware

TK-2187 — DELTA INC — SEVERITY: CRITICAL

SOP steps

- 8. Default inputs — CrowdStrike token, SCCM key, VLAN config
- 9. API call (MCP tool: edr_endpoint · isolate_host) — isolate WKST-042
- 10. Agent — forensic analysis: scan artefacts, map lateral movement
- 11. Conditional — Lateral spread? Yes: API call (MCP tool: network_manager · block_vlan)
- 12. Router — remediation type: patch → API call (MCP tool: sccm_manager · deploy_patch)
- 13. Human task — SOC sign-off
- 14. End

Key decision-point JSON

Decision point	Sample JSON
Agent forensics output	{ "lateral_spread_detected": true, "affected_vlans": ["VLAN-14"] }
SCCM patch output	{ "patch_kb": "KB5034441", "status": "installed", "reboot_required": true }
SOC sign-off	{ "decision": "approved", "completed_by": "analyst_raj.kumar" }

7.3 Client C — Gamma Bank: Data Exfiltration

TK-3391 — GAMMA BANK — SEVERITY: CRITICAL

SOP steps

15. Default inputs — DLP API key, HR token, legal hold URL
16. API call (MCP tool: dlp_manager · get_transfer_log) — pull p.chen session log
17. Agent — transfer classifier: destination, file types, volume, intent
18. Conditional — Confirmed exfil? Yes: API call (MCP tool: iam_manager · suspend_user)
19. Human task — compliance officer: approve legal hold?
20. Conditional — Legal hold needed? Yes: API call (MCP tool: legal_manager · create_hold)
21. Notification — email CISO and legal team
22. End

Key decision-point JSON

Decision point	Sample JSON
Agent classifier output	{ "exfil_confirmed": true, "destination_category": "tor_exit_node", "confidence": 0.98 }
IAM suspend output	{ "user": "p.chen", "iam_status": "suspended", "tokens_revoked": 3 }
Legal hold output	{ "hold_id": "LH-2025-441", "status": "active", "expires": "2026-05-30" }

Section 8: JSON Context Reference & Platform Constraints

8.1 Context Key Namespaces

Node	Shape · Colour	Ports	Rationale
\$.input.*	\$.input.ticket_id	Webhook or scheduler	Injected from request body or static_input
\$.defaults.*	\$.defaults.me_api_key	Default inputs block	Available to all downstream nodes
\$.loop.*	\$.loop.index · \$.loop.item	Loop node	Per-iteration state injected into body context
\$.wait_meta	\$.wait_meta.exit_reason	Wait node	Appended after timer or process wait completes
\$.mcp_result	\$.mcp_result.device_id	Connector MCP mode	Full tool response from AutoX MCP server
\$.mcp_meta	\$.mcp_meta.duration_ms	Connector MCP mode	Execution metadata: duration, retries, status
\$.loop_results	\$.loop_results[0].status	Loop Done port	Collected iteration outputs array
\$_branch	\$_branch: true	Conditional node	Records which branch was taken
\$_route	\$_route: CLIENT_B	Router node	Records which route was matched

8.2 Platform Constraints

Node	Shape · Colour	Ports	Rationale
Max JSON payload per node hop	5 MB	—	ERR_FLOW_003 if exceeded
Max flow context size	10 MB cumulative	—	ERR_FLOW_003 if exceeded
Max concurrent flow runs	10 per flow default	Configurable	ERR_FLOW_007 if exceeded
Webhook max body size	1024 KB default	Configurable up to 100 MB	ERR_WH_003 if exceeded
Scheduler minimum interval	1 minute	—	Validation error on save
Agent max steps	50 hard ceiling	—	ERR_AG_001 when hit

Loop max_iterations	100,000 hard ceiling	—	ERR_LP_002 when hit
Process call timeout default	300 s	Configurable up to 3600 s	ERR_PC_002 when exceeded
Human task SLA default	60 min	Configurable up to 30 days	ERR_HT_002 on expiry
Wait node max duration	30 days	—	ERR_WT_003 if exceeded
MCP tool call timeout	30,000 ms default	Configurable up to 300,000 ms	ERR_CN_002 when exceeded
Router max branches	50 named routes	—	Validation error on save
Notification max recipients	200 per send	—	ERR_NO_002 if exceeded
Flow run log retention	90 days	—	Purged automatically